

Users' involvement may help respect social and ethical values and improve software quality

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Abstract Global informatization brings potential threats. It is possible to reduce them if software developers are aware of those threats and if customers insist on eliminating them. The survey described in the paper shows that future programmers are aware of numerous threats associated with global informatization. Many threats may be reduced by users' involvement in the software process. Software developers should learn and develop cooperation with users, as emphasized in agile methodologies. It is a task for academic teachers to promote collaboration between software authors and users, to show how to organize it, and to convince programmers that it is valuable. Users' involvement requires changes in the software life cycle and in the set of established processes. Such changes, in the form of additionally established processes and phases in the software development cycle, are identified in the paper. One recommended process is the software product assessment intended to elicit feedback from users. Users' involvement may help to respect ethical and social values and to produce software adapted to user expectations.

Keywords Threats related to extensive informatization · Social threats · Software quality · Users' involvement · Feedback from users · Primary and supporting processes in software development · Agile approach

1 Introduction

Software products, including expert systems built for public organizations, are interactive in nature and require semantic integration of data and processes maintained in various institutions providing services for citizens. Direct and indirect users often complain of poor quality of the delivered software systems and their data. Citizens have to visit a public organization several times to fix their official matters (private and business affairs), as it was before the informatization era. The existing procedures require citizens to deliver a lot of paper documents, and to do it many times. As far as data correctness is concerned, a lot of mistakes are met in databases because of insufficient data validation. And it should be remembered that the correctness of the maintained data determines the quality level of services rendered to citizens by public organizations.

Informatization brings potential threats. The inconvenient procedures, occasional system failures, and maintained false data are not the only examples of threats in social life. The list of threats is much longer and includes, among others, health risks, unethical attitudes and actions, and damage of social relationships. The questionnaire survey, carried out among computer science students of PUT and reported in Section 2, shows students' awareness of various threats associated with extensive informatization on different scales—an individual scale, a country-wide scale, and a global scale of informatization.

Respecting ethical and social norms and ensuring software quality are not granted by any decree. Software companies are first of all concerned with productivity. Hard methodologies, which are dominating on the market, are based on detailed definitions of processes, various rigorous plans, and precise document templates. All these elements help to manage the software process as long as bureaucracy

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(i.e. the number of procedures and documents required in the software process) does not exceed a reasonable level. Software quality in hard methodologies is equivalent to following rigorous plans and to keeping to detailed definitions of processes. Software developers usually act in isolation from users but they more and more have to take into consideration their customers' opinions. Customers and users may and should desire companies to respect ethical and social norms in software production and ensure software quality as understood by the customers and users. New software principles and tools are required to allow the production of high-quality applications for global computing, cf. the document of the GC Strategy Group (The GC Strategy Group 2003).

One strategy to improve software quality is to provide continual feedback from users on software products, especially on expert systems and software products built for public organizations. Software customers and users are more and more frequently asked to provide services for themselves using Internet-based solutions. Therefore users should be able to influence the interactive systems development (Mattsson 2009). Collaboration with users and direct communication are characteristic for agile methodologies. They have been developed in response to bureaucratic and unsuccessful methods in software production. Some software producers, especially in Scandinavia, cooperate with users. In fact, they employ different approaches to user involvement although users' position is not equal in various organizational cultures (Iivari 2004).

The main idea emphasized in this paper is to incorporate users' involvement in the software process and to make users the subject of it to build high quality software products. To realize this idea several additional primary and supporting processes are required in a software life cycle. The processes recommended to provide ongoing successful cooperation of software companies and users are presented briefly in Section 3. Several internal loops of feedback are provided in each development cycle, representing active involvement of users' representatives in various working teams, meetings, brainstorming sessions, test design and supervision, software inspections (where users are co-authors of control lists of inspections), etc.

Software product assessment, intended to elicit the feedback from users, constitutes the external loop of the software life cycle. The author's research shows that even if cooperation with users is limited to software product assessment, it brings positive results if a wide spectrum of users are consulted. In general, users are eager to present their ratings of a given software product and to formulate suggestions for improvements. An instrument to assess software quality is a questionnaire survey, possibly involving all users, described in Section 4. The analysis of the obtained results allows to specify software improvements.

This form of cooperation is recommended when starting to apply agile methodologies.

Agile approaches allow more freedom of action but at the same time require special competencies of the software factory staff, different from and/or additional to those desired in hard methodologies (Turner and Boehm 2003). So it is necessary to educate software designers to cooperate with users. Some suggestions on how to prepare software producers for such collaboration are given in Section 5, devoted to conclusions.

2 Students' awareness of threats related to extensive informatization

The information revolution brings a lot of well-known benefits but at the same time it may threaten human values. Human aspects of extensive informatization on the global scale seem to be overshadowed by technological innovations. Thus the research does not concentrate only on the technological progress but its ethical and social aspects are discussed (the ETHICOMP conference series provides an international forum for such discussions). But awareness of the importance of interpersonal relations is also crucial for software engineers. Cooperation of software developers with users makes it easier to consider the development of ICT (Information and Communication Technologies) as a facilitator of social interactions and environmental wellbeing.

A questionnaire survey entitled "Social and ethical threats related to extensive informatization" was conducted to assess computer science students' awareness of threats to ethical standards and social norms. Students who took part in the survey had not attended extra classes concerning ethics or ethics in computing. But they were not completely ignorant of social aspects in computing—students had already discussed selected problems concerning the authors' law, protection of the Polish language by law, legal aspects of contracts regulated by the civil code (including a contract for a specific task and a fee-for-task agreement), and legal protection of personal data. Topics concerning ethics in computing have since been incorporated in the course of software engineering given by the author.

The survey was directed to all third year computer science students of the Faculty of Electrical Engineering, PUT. As an encouragement, students had complete freedom to articulate their indications of threats—the questionnaire was based on open questions. To avoid focusing only on the individual level, the potential threats were a priori divided into three categories, related to an individual scale (like health problems, intellectual development, and psyche), a country-wide scale, and a global scale. In the survey, students were asked to identify threats caused by

informatization. Respondents had to make a list of threats which they considered most serious. In each category, 1–5 items were expected. Additionally, respondents were asked to say whether they spoke from their personal experience (P) or if they presented “second hand” information (F) that they heard, read or met somewhere. Out of 84 registered students as many as 72 students filled in the questionnaires on May 20th, 2008. It took them ca 15 min to do it. Besides, two students saw no threats at all.

The respondents named altogether 269 items in the first part of the questionnaire (3.74 on average). Many respondents gave the same threats in their questionnaires so it was possible to work out a ranking of potential threats with reference to individuals. Serious health risks occupy the first position. The top twelve items are listed in Table 1. Items in the table retain original wording given by the students.

Other threats indicated by respondents were:

- Feeling of being overwhelmed with technical equipment
- Dehumanization; e.g. humans being reduced to their Personal Identification Numbers
- Problems with access to required information in critical or emergency situations (e.g. to medical documentation)
- Fear caused by insufficient security of personal data, including medical data
- Permanent control of our lives, as in the case of Orwell’s Big Brother

- Digital gap: poor or uneven distribution of computers, unequal availability of required information, high prices of Internet connections, problems with using computer and electronic equipment
- Nervous breakdown of a person, following the publication of false and/or offensive material about his/her in the Internet
- Mobbing and/or being plagued with unwanted messages (spam) via the Internet
- Degeneration of childhood (playing with machines instead of people)
- Doing unnecessary and/or stupefying work using document templates only
- Pupils’ reluctance to learn because “everything” is in the Internet
- Uncontrolled sale of illegal drugs and other products via the Internet
- Plagiarism, cheating at school made easy by the Internet, etc.
- Abuse (of children, gullible people, etc.) by pedophiles or fraudsters.

In many cases concerning all three categories, the respondents failed to point out what threats they faced and/or which problems they met themselves, and which were known to them from other sources. Students indicated first-hand experience in 52 cases and other sources in 104 cases. In 113 cases the source was not mentioned. Health

Table 1 Potential threats caused by informatization with reference to individuals

Place	Kind of threat	Source	Votes
1	High risk of illness (eyes, bone joints, backbone, nervous system) and stress caused by sitting permanently indoors in front of a computer screen; generally weak physical condition	P, F	53
2	Getting lost in a virtual world: blurred boundaries between real and virtual life (moving life to the Net), equating oneself with virtual game characters, deterioration of interpersonal relationships, more time spent with a computer than with real persons, growing shyness, problems in real life, participation in a virtual life instead of reality	P, F	29
3	Weak intellectual development: thoughtless acting, clicking a mouse only, poor hand-writing, loss of literacy, decreasing creativity, deterioration of logical thinking, copying ready solutions and using schema from the Internet, poor imagination, passive instead of active lifestyle	F	26
4	Social alienation: escape from the real society, degeneration of interpersonal communication, weak communicativeness, lack of empathy	P, F	23
5	Internet addiction and/or game addiction	P, F	22
6	Easy access to unethical materials (especially destructive for children), bad influence of violent computer games on an immature psyche	P, F	20
7	Decreasing number of direct contacts with other people, lack of friends, isolation from old friends	P	13
8	Hiding own personality and creating a virtual one just to demonstrate bad emotions, absence of mental blocks, decreasing responsibility for words and actions	P, F	8
9	Disappointment, loss of trust, and/or fear caused by maintained erroneous or uncertain data, false data found in the Internet, loss of data caused by computer failure, illegal interception of personal data	F	7
10	Lack of individual cultural development, decrease of direct participation in cultural events, reading blogs instead of books	F	6
11	Creeping consumerist style of living, excessive impact of electronic media, using information technologies to manipulate people, often against their interests	P	5
12	Inflow of nonessential information; problems with searching for required information, no difference between valuable information and rubbish	P, F	3

problems with eyes and backbones (caused by sitting motionless in front of a computer screen) were the most often mentioned personal troubles. Students also experienced social alienation, lack of true friends, and exposure to violent and perverse, sexually explicit graphic material and false data found in the net.

A relatively smaller number of items, namely 217 threats, were pointed out in the second part of the questionnaire. So it gave 3.01 items on average. The results related to the country perspective are gathered in Table 2 in the decreasing order of obtained votes. The two most often mentioned issues were insufficient data security and naive belief that new technology may bring only benefits.

Several respondents also emphasized the following factors:

- Callousness of machines which treat a person like a row in a table or a number in a database
- Anonymity of users who demonstrate lack of responsibility for their words and opinions (stupid, thoughtless, ill-judged, false)—some people may be hurt by that
- Growing bureaucracy—too many data items are demanded by administration and maintained in databases
- The fact that non-typical cases are not supported by computer systems and thus some applications of indirect users may not be fixed at all
- Corporate control over clients
- Monopolization of nation-wide information systems—it may lead to corruption
- Increasing problems with electronic waste (unnecessary equipment and data)
- Computer viruses spreading across the world
- Difficulties caused by the incompatibility of computer systems and data

- Growing consumption of energy due to increasing amount of electronic equipment; power shortages are imminent
- The fact that legal regulations and courts do not keep up with the changing world.

In the country-wide category 26 threats were personal experiences of the respondents and 97 items were known to them from elsewhere. No source was indicated in the remaining 94 cases. Some students blamed financial problems or even poverty in their families on automation of the office work and transformation of management in general, resulting in decreasing number of work places. Thus they indicated that poverty is a threat. Unauthorized access to protected data, digital gap between the more and the less developed parts of the country and impersonation of other people were given as personally known threats on the country-wide scale. The respondents also pointed out to exposure to undesired materials from the social and ethical point of view. Some students experienced the effects of incorrect data maintained in the public system and the painful consequences of the information system failure.

All together, 234 items were given in the last part of the questionnaire, devoted to potential threats on the global scale. It gave 3.25 threats on average. The respondents did not see much difference between the threats in the country perspective and the global perspective. They mentioned few new issues in this category. The results are presented in the decreasing order in Table 3.

28 threats on the global scale were already experienced by students and 110 were known from other sources (no source was indicated in 96 cases). In this category young respondents experienced piracy (using illegal software or being harmed by intellectual theft), leak of confidential

Table 2 Potential threats caused by informatization on a country-wide scale

Place	Kind of threat	Votes
1	Lack of privacy—easy access to other people's personal data, possible unauthorized access to protected data, lack of individual control over personal data in information systems, resale of personal databases	44
2	Boundless trust in computer systems—lack of access to computer resources in the case of a power failure (from the local to the country-wide scale), lack of alternative procedures in the case of software failures or disasters, possible total paralysis of a country, including airports, banks, etc.	23
3	E-burglary and e-theft (including banking data), cyber crime, supporting criminal gangs	17
4	Digital gap: growing social differences, weak communication network in under-populated areas, elderly people's problems with electronic systems	14
5	Disappearance of tradition, declining national pride, cosmopolitanism, cluttering mother tongue with foreign words	12
6	Being under permanent surveillance; possibility of locating a person at any time	11
7	Possible impersonation of other people, identity theft	11
8	Piracy, theft of intellectual property	9
9	Problems with erroneous data and software failures	9
10	Manipulating and stupefying people by using deceitfully selected information and falsifying data	7
11	Decreasing number of jobs because of total informatization	6
12	Espionage and industrial spying	5

Table 3 Most frequently named threats on the global scale

Place	Kind of threat	Votes
1	Lack of privacy, unauthorized access to personal data	31
2	Breaking into databases (including military systems), resale of stolen data, fraud on a large scale, money laundering	31
3	Possible chaos in the case of a global failure of computer networks	19
4	Digital gap between the rich and the poor societies	13
5	Piracy	13
6	Possible impersonation of other people, identity theft	12
7	Manipulating people, lack of responsibility, circulating malicious software and nasty data	11
8	Cyber-terrorism, cyber-attacks	8
9	Espionage	6
10	Surveillance on a large scale, acting as Orwell's "Big Brother"	6
11	Decline of local cultures	5
12	Monopolization of the global market	5

information to the Internet, circulation of false information, and dependency on unreliable system. They erroneously identified virtual reality with modernity and joined the widespread pursuit of wealth. They also observed a generation of Internet kids who have no true childhood.

3 Emerging methodologies and suggested subprocesses of the software process

Process-oriented hard methodologies were born in the military context and they are still reported as successful in military-related areas, such as arms production and aircraft industry (Boeing, Lockheed Martin), and in the branches of computing and telecommunication (Motorola). During the 2005 Software Engineering Process Group (SEPG) conference, the armament sector was represented, among others, by Hewlett-Packard, IBM, Motorola, and Siemens. Currently, hard methodologies are also applied to produce software systems for civilians. But hard methodology procedures, including those following the recommendations given in the ISO 9000 and/or the models of the CMMI, do not guarantee high or even sufficient quality of the software product from its users' point of view. Specifically, they often fail when applied to software built for public organizations. Planning and discipline form the foundation for success in many areas, also in software production. But discipline does not mean that efforts always bring results expected by product users.

It has been shown that the promoted users' involvement in software development improves the analysis of user needs (Kujala 2008). Cooperation with users has been widely accepted in agile approaches (Agile_FDD 2004; Ambler 2006; Beck 2000). Two of the twelve principles expressed in the Agile Manifesto concern collaboration with users and say that *business people and developers*

must work together daily throughout the project and that *the most efficient and effective method of conveying information to and within a development team is face-to-face conversation* (Manifesto 2001). The emerging agile methodologies encourage customer-developer collaboration (Highsmith 2004), combine human efforts and technology by making use of people's competencies (Cockburn 2005) and prepare participants for collaboration (Tabaka 2006). Specified practices are recommended to guide a project in an agile approach (Martin and Martin 2007).

Growing popularity of agile methodologies encourages software developers to follow principles listed in the Agile Manifesto, to respect the Software Engineering Code of Ethics (Gotterbarn 1999) and to include ethical issues in the software project management process (Rogerson and Gotterbarn 1998). Thus, agile methodologies may be found to be more human-oriented than other methodologies. But there are so far some serious concerns about agile methodologies. Namely, in larger teams the software process may become chaotic and inefficient. We may expect agile projects to work best when small teams (less than 20 team members) are involved (Beck 2000; Cockburn 2005).

The cooperation with users should be neither declarative nor spontaneous only. On the other hand, we must prevent the software process from falling into chaos. Balancing agility and discipline in the software process is a serious challenge. It is necessary to find a middle ground between hard and soft methodologies (Boehm and Turner 2004; Begier 2005). In the conventional development cycle, the role of all stakeholders considerably increases (Biffl et al. 2006). In turn, agile approaches acknowledge the need for some defined practices (Ambler 2006; Beck 2000).

Most threats mentioned in the second section are results of global informatization supporting the global economy. Although software authors cannot eradicate social evil and are not responsible for all reported threats, the recommen-

ded cooperation of software developers with users results in better awareness of those threats in users' communities. Then remedy may be found for some identified threats. Ethical and social aspects of software applications may be considered during an additional phase of software development cycle (Begier 2002). Ongoing cooperation with users' representatives makes it possible to learn the point of view of various users and to improve a product according to particular software measures and user expectations.

In order to reduce/eliminate potential threats and cooperate with users, it is necessary to shift the emphasis in the software process from productivity to product quality. In this context, the *objectives of the software process* singled out by the author are the following:

- focus on product, starting from a software product vision,
- focus on people in both supplier and purchaser organizations
- high level of users' satisfaction from the software product
- innovative solutions and product appropriateness to business and users' expectations.

Software development is based on the iterative-incremental model. Frequent small increments of software functionality make possible responding to changes in customer requirements. Product usability improvements are based on the feedback from users. A framework for integrating usability practices in the software process is needed (Ferre et al. 2005). The established software process determines the quality of the software product and its continual improvement.

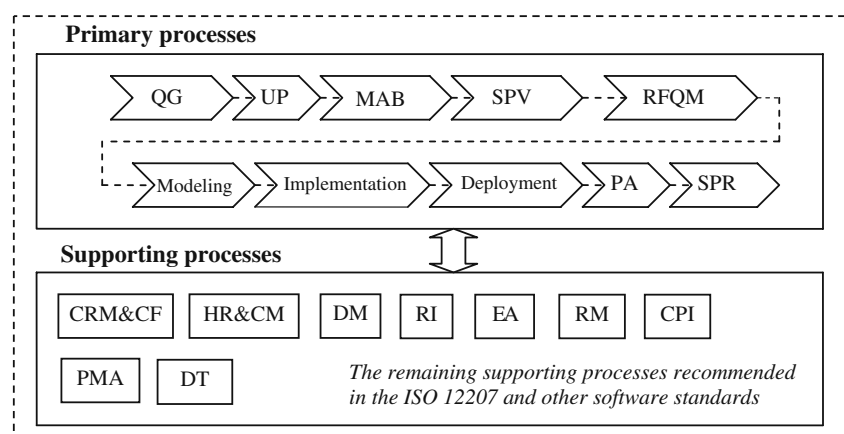
In business engineering all processes are divided into the *primary processes* and the necessary *supporting processes*. The former processes lead to developing a specified product version and its delivery to a customer. The latter processes support the correct performance of the primary processes. This also refers to software factory. The set of

conventionally required processes has been extended as shown in Fig. 1. Several additional processes, either primary or supporting, are introduced to ensure the users' involvement in software development. Especially, the subprocesses named QG, PA, and SPR are the author's proposal. Some processes listed in Fig. 1 have been changed to accommodate the needs of cooperation with users (Begier 2007). The graphical notation developed for the *ARIS Quickstep for SAP R/3* has been applied to present a partially ordered sequence of primary processes as the subsequent pieces of a cut-up tape. Transitions on the process map symbolize transfers of an added value. The order of processes, symbolized by a dashed line, is not definitely determined because of possible recurrences resulting from users' opinions and quality assessment after each development phase.

The following primary processes are recommended by the author to ensure cooperation with users:

- QG (*Quality of Goals*)—specification of goals of a given project and their quality analysis
- UP (*User Profiles*)—specifying users' profiles on the base of a questionnaire survey
- MAB (*Modeling & Analysis of Business processes*)—modeling and analysis of business processes to be supported by the software product under development; the aim is to avoid yesterday's solutions being built into a new system
- SPV (*Software Product Vision*)—development of a software product vision, including how the product will be used; it is typical for the agile approach (Highsmith 2004),
- RFQM (*Required Features and their Quality Measures*)—specifying required software features, including criteria and measures of their quality
- *Modeling* based on close cooperation with users
- *Implementation* including users' involvement in test planning, design of test cases, and then providing users' supervision of testing

Fig. 1 A map of primary and supporting processes necessary to produce one version of a software product in the recommended approach



- *Deployment*—including trainings arranged according to the user profile
- *PA (Product Assessment)*—software product assessment conducted by possibly all users or by a representative group of users in the case of software products used on a mass scale, followed by analysis of potential threats referred to the quality in use
- *SPR (Specification of a Product Refinements)*—working out a specification of product refinements based on the results of a product assessment (Begier and Wdowicki 2006).

The quality level of primary processes depends on correct realization of supporting processes. Activities of one process interweave with activities undertaken in other processes and/or overlap these activities. The suggested list of supporting processes is as follows:

- *CRM&CF—Customer Relationship Management & Continual Feedback* from users on a software product and indirectly on a software process
- *HR&CM—Human Resources & Competency* in a software company
- *DM—Document Management* limited to the specified needs, like test plans and reports
- *RI—Reviews & Inspections* in which users' representatives are involved
- *EA—Ethics Assurance* in a software project
- *RM—Risk Management*, including poor cooperation with users as a source of risk
- *CPI—Continual Process Improvement*, including proposals given by the staff
- *PMA—Process Measurement & Assessment*, including maturity of user-developer relationships
- *DT—Development of Technology* oriented to user's satisfaction from a product.

The implementation of the processes listed above determines the quality of the entire software process and,

consequently, the quality of the software product. Especially, the Customer Relationship Management, as a part of the CRM&CF system, should be extended to manage contacts with various categories of users who participate in everyday activities of software developers. The significance of particular subprocesses depends on the type of software product. The performance of one process potentially determines the quality of other processes.

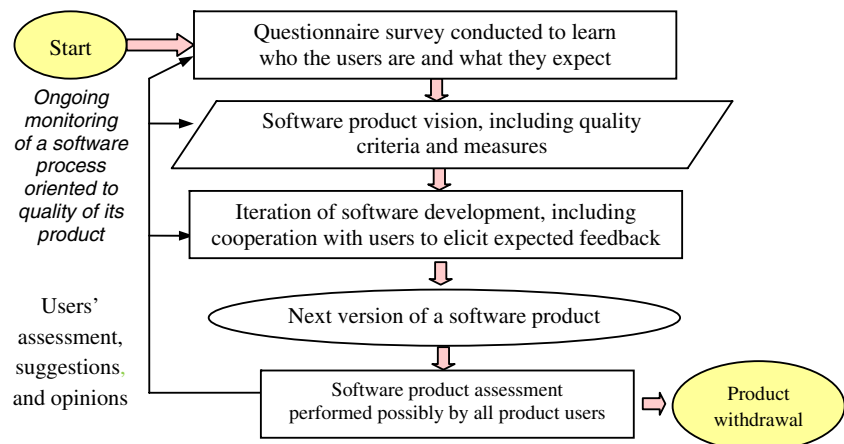
4 Software product assessment as an external feedback from users

Conventional phases of software development no longer make sense in the agile approach (Ambler 2005). To provide continual feedback from users on the software product, additional phases are introduced in the software life cycle, as illustrated in Fig. 2. The goals are to ensure the users' high satisfaction with a product and to ensure that ethical and social norms are respected. It means that all realized processes are in fact (not declaratively only) focused on these goals and additional activities are introduced to make it happen. First, users' profiles are built and users' expectations are learned. Then the product vision may be jointly developed by software producers and user representatives, as it is usual in agile methodologies. The required feedback is not limited to working out the specification of requirements but it should cover the entire software life cycle.

The working software is the primary measure of progress in software development. In the author's proposal, the essential phase closing each development cycle is the product assessment performed possibly by all software users. Managerial and marketing techniques, such as questionnaire surveys, are adapted to that purpose, becoming also practices of software engineering.

The author's research shows that the periodical assessment of each deployed software version by its users brings

Fig. 2 Software life cycle based on continual feedback from users



valuable information (ratings, suggestions of improvements, and opinions) for software developers. The author has applied the approach outlined above to develop a software system for civil engineering (Begier and Wdowicki 2006). The authors proposed the content of questionnaires and conducted five iterations of the product assessment. Subsequent versions of the product were developed in 2002–2007 taking into consideration the results of the conducted surveys. The authors notice that the level of users' satisfaction from the product is currently much higher than at the beginning.

The described approach may be applied to evolutionary development of business and expert software systems, and especially those designed for public organizations. The question is whether software users are ready to participate in software improvement. The surveys by questionnaire conducted by the author in four organizations show that many users are eager to indicate software features to be improved. But their enthusiasm may wear off if their ratings and suggested changes are not reflected in further software development.

5 Summary and conclusions

The following issues are discussed in this paper:

- awareness of potential threats posed by new information and communication technologies among today's students—software developers and software users of tomorrow
- additional processes and phases to include cooperation with users in software development
- quality assessment, performed by software users, of each deployed software version
- necessity to incorporate ethical and social aspects of information technologies in computer science programmes
- additional responsibility of all academic teachers with respect to developing awareness of potential threats in the whole society.

The aim of the described (and recommended to conduct in other universities) survey is to learn the degree to which the potential threats are known among software authors and users. The reported questionnaire survey shows that today's students and future programmers are aware of at least some potential threats of different nature associated with global informatization. Many indicated items were based on students' own experience. In this sense, the results of the questionnaire are positive. In the respondents' opinions, health risks are the greatest threat. Lack of privacy, including lack of human control concerning data security in information systems, has a leading position among threats on the country-wide scale and on the global scale.

Many reported threats may be reduced or even eliminated if considered in the software process. In the country perspective, many threats may be averted if users are involved in the software process, and if they are encouraged to apply the jointly designed techniques and undertake required activities. It refers to solutions concerning data protection, safety of computer systems, careful design of information system procedures, education of adults, essential legal regulations, etc. The considerations presented in this paper refer especially to software applications, including expert systems, supposed to improve activities of public organizations from the point of view of direct and indirect users.

However, to ensure the desired developer-user collaboration, additional primary and supporting processes must be introduced in the software process. Several processes of this kind are identified in this paper. One of them is the software product assessment intended to elicit feedback from users. Practice shows that users' satisfaction with a given software product after periodical software assessment and ensuing improvements is higher than at the beginning. In turn, ethics assurance should be incorporated in software development to prevent threats from materializing. Organizational and computing support is required to make the ongoing cooperation with users' representatives possible and successful.

Results of ongoing cooperation with users depend on competencies of the personnel of a software factory. Competencies of the staff seem to be a critical component of agility (Turner and Boehm 2003), allowing the staff to respond to continual changes of external environment, including users' growing expectations, progress in technology and business conditions. The competencies required in agile methodologies are much more diverse than in the case of hard methodologies. The Human Resources & Competency Management process is intended to support the cooperation with users. There is a need to build competency profiles for essential professional roles and then to use them in internal and external recruitment procedures. The competency profile is a set of competencies required to perform a given role. Specification of each competency profile should include not only technical skills but also personality features helpful in face-to-face relationships with users. All those activities help develop *a culture of excellence*.

The quality of technical solutions definitely depends on the applied methodology, on social honesty of decision makers, and also on the awareness of potential threats in societies. Respecting ethics in computing requires incorporating ethics in the educational process and providing methodology specified for this purpose (Huff and Martin 1995). The role of academic teachers is to include ethics in computing in the education of future software designers and decision makers. The subjects dedicated to ethical and social problems should become an integral part of a software engineering course—it is crucial to make students

aware of social, ethical and cultural impacts of information technology. Human-oriented methodologies, including agile ones, should be recommended to present and future software engineers. Then software developers will apply the described new elements in software production with full conviction.

If threats caused by informatization on a global scale are to be reduced or eliminated, it is necessary to raise the awareness of these threats and to emphasize them in the education of the entire society. Some practices introduced by informatization seem initially advantageous but they may bring completely undesired effects. So far software products support, first of all, activities which are effective, calculative, predictable and repeatable—and it sounds reasonable at first glance. But this tendency may lead to unification of everything on a global scale, to ignoring local cultures, and to passing over all cases which do not match the routine procedures. Thus this tendency supports directly the so called *McDonaldization of society* (Ritzer 1996). Software investors and developers are expected to predict, analyze and manage this problem, too.

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